

DEV_IMPL_GUIDE - FE-APP-03 Contractor Technical Mobile App

1. Implementation Objective

Implement the contractor technical mobile app for assigned work orders, offline execution, evidence capture, equipment scan/validation, job finalization, and field issue escalation.

2. Start Here

Read these first:

1. FE-APP-00-Frontend_Architecture_and_Shared_UX_Foundation-v1.0.md
2. FE-APP-03-Contractor_Technical_Mobile_App-v1.0.md
3. DD_WO-01-Work_Order_Module-v1.0.md
4. DD_WO-01-FRAMEWORK-v1.0.md
5. DD_OSR-INSTANCE-01-Equipment_Instance_Registry-v1.0.md
6. DD_OSR-01-Stock_Chain_Engine-v1.0.md
7. DD_EM-02-Contractor_and_Staff_Registry-v1.0.md
8. FOUNDATION_FILE_STORAGE.md

3. Build Order

1. Implement authenticated job list from WO.
2. Implement job detail, checklist, notes, and field status transitions.
3. Implement offline work packet download for assigned jobs.
4. Implement evidence capture and upload through file storage.
5. Implement equipment serial scan and validation through OSR before finalization.
6. Implement WO finalization and sync retry.
7. Implement field issue creation/update through TCK when needed.

4. Offline Rule

Offline mode may store assigned job packets, notes, photos, and pending finalization commands. It must not invent equipment validity, subscription state, ticket status, or stock ownership. Those are confirmed by owning APIs during sync.

5. Tests

- Job list and claim.
- Offline evidence capture.
- Sync retry and idempotency.
- Invalid equipment serial.
- WO finalization with file upload.
- Field issue creates ticket.

6. Screen-To-API Wiring

Screen	Owning module API	Calls
Login/contractor context	FOUNDATION_AUTH, EM-02	OIDC login, contractor/team/skill context.

Screen	Owning module API	Calls
Assigned jobs	WO-01	assigned work-order list, claim/accept where allowed.
Job detail	WO-01	work-order detail, appointment, checklist, customer/account/homepass summary.
Equipment scan	OSR-INSTANCE-01, OSR-01	validate serial/MAC, check ownership/location/assigned stock.
Evidence	FOUNDATION_FILE_STORAGE	upload photos, signatures, test results.
Job finalization	WO-01	finalize, fail, reschedule, request help.
Field issue	TCK-01	create/update ticket linked to work order.

7. Offline Work Packet

Download an assigned job packet before field work:

```
type WorkPacket = {
  workOrderId: string;
  accountId: string;
  customerId: string;
  homepassId?: string;
  appointmentWindow: { start: string; end: string };
  jobKind: "INSTALL" | "REPAIR" | "SWAP" | "RECOVERY" | "SURVEY";
  checklist: Array<{ code: string; label: string; required: boolean }>;
  expectedEquipment?: Array<{ skuRef: string; quantity: number; serialRequired: boolean }>;
  allowedActions: string[];
  downloadedAt: string;
};
```

Offline packet data is a work aid, not source of truth. On sync, the app must revalidate work order status and equipment state before finalization.

8. Flow - Finalize A Work Order

1. Contractor opens assigned job from WO-01.
2. App downloads packet and lets contractor capture notes, checklist answers, photos, GPS, and serial scans offline.
3. For each equipment serial, app validates online when possible:

```
GET /api/equipment-instances/lookup?serial=ONT123456
Authorization: Bearer <jwt>
```

4. App uploads evidence files:

```
POST /api/files/upload-sessions
Content-Type: application/json
```

```
{
  "purpose": "WORK_ORDER_EVIDENCE",
  "ownerModule": "WO",
  "ownerRef": "wo_01HZ"
}
```

5. App submits finalization:

```
POST /api/work-orders/{work_order_id}/finalize
Idempotency-Key: wo-finalize:{work_order_id}:{localAttemptId}
Content-Type: application/json
```

```
{
  "outcome": "COMPLETED",
  "completedAt": "2026-06-01T14:30:00+03:00",
  "checklist": [
    { "code": "SIGNAL_TEST", "value": "PASS" }
  ],
  "equipment": [
    { "serial": "ONT123456", "action": "INSTALLED" }
  ],
  "evidenceFileRefs": ["file_01HZ_PHOTO"],
  "gps": { "lat": -1.2921, "lng": 36.8219, "accuracyM": 12 }
}
```

6. WO-01 validates status and records finalization.

7. Downstream WO/OSR/FUL events update equipment and service state. The app waits for confirmed API response; it does not locally mark inventory as installed.

9. Sync And Conflict Handling

Conflict	UI handling
Work order cancelled while offline	Stop sync, show cancelled state, keep local evidence read-only for supervisor review.
Serial belongs to another account/location	Block finalization and require replacement serial or dispatcher override where policy allows.
Required evidence missing	Keep command local in FAILED_CONFLICT; highlight missing items.
Upload succeeds but finalization fails	Keep uploaded file refs and retry finalization with the same idempotency key.
Duplicate finalization retry	Show existing WO finalization result returned by WO-01.